

Assessment 03 - Raima Khan

Q1. What is Sequence Analysis?

Sequence analysis is the process of using the computational, statistical and comparative methods to examine the DNA, RNA or protein sequences in order to extract useful biological information about the composition, structure, function and evolutionary relationships. It transforms raw seq. data generated by sequencing technologies into interpretable biological knowledge.

Importance in Bioinformatics

Seq. analysis forms the foundation of modern molecular biology and genomics. Raw seq. data produced by Sanger sequencing or Next-Generation Sequencing (NGS) has no biological meaning on its own, it must be analyzed to reveal genes, regulatory elements, mutations and evolutionary context. Seq analysis bridges the gap between raw genetic code and functional biological understanding and underlies applications ranging from disease diagnosis and drug design to crop improvement and the evolutionary research.

Four Major Applications

1. Gene identification & annotation:

Locate open reading frames (ORFs), exons, introns, promoters and the regulatory elements within the genome using tool such as BLAST

2. Comparative genomics & phylogenetics:

Compare the seq across species or individuals to study the evolutionary relationships, construct phylogenetic trees and identify conserved and functionally important regions.

3. Mutation & variant detection:

Identify the SNPs, insertions, deletions or the structural variants linked to disease or agronomically important traits.

4. Molecular marker development (MAS):

In the plant breeding, seq analysis identifies polymorphic regions (SSRs, SNPs) used as the molecular markers for Marker-Assisted Selection of desirable traits.

Q2. GC and AT Content

GC content is the percentage of nucleotide bases in a DNA (or RNA) sequence that are either guanine (G) or cytosine (C):

$$GC\% = [(G + C) / \text{Total nucleotides}] \times 100$$

AT content is the percentage of bases that are adenine (A) or thymine (T) (uracil in RNA):

$$AT\% = [(A + T) / \text{Total nucleotides}] \times 100$$

Since every base is classified as either GC or AT, the two values always sum to 100%: $GC\% + AT\% = 100\%$.

Why These Parameters Matter

The GC and AT content matter across several areas of the genome analysis and molecular biology. In terms of thermal and chemical stability, the G–C pairs form three hydrogen bonds compared to only two in the A–T pairs, so the higher GC content raises the DNA melting temperature and increases overall double-helix stability. This same principle underlies PCR primer design, where the GC content determines primer annealing temperature; primers are typically designed with 40–60% GC to ensure specific and reliable binding. The GC content also serves as a genomic signature for organism and genome characterization, for example, *Streptomyces* has a GC content of around 70%, while *Plasmodium falciparum* has only about 20%, differences that can reflect environmental adaptation. It is likewise useful for gene density and structure prediction, since the GC-rich regions often correspond to gene-dense, exon-rich areas while the AT-rich regions are more commonly associated with introns and non-coding DNA. Finally, GC/AT content has practical implications for sequencing and cloning, as extreme GC or AT content can cause PCR amplification difficulty, sequencing errors and secondary structure formation.

Q3. Scientific Problem — TP53 mRNA (NM_000546.6)

Source: Homo sapiens tumor protein p53 (TP53), transcript variant 1, mRNA RefSeq accession: NM_000546.6 (NCBI Nucleotide database).

NIH National Library of Medicine
National Center for Biotechnology Information

Nucleotide Nucleotide Advanced

FASTA Send

Homo sapiens tumor protein p53 (TP53), transcript variant 1, mRNA
NCBI Reference Sequence: NM_000546.6

GenBank Graphics

>NM_000546.6 Homo sapiens tumor protein p53 (TP53), transcript variant 1, mRNA
CTCAAAAGTCTAGAGCACCCTCCAGGGAGCAGGTAGCTGCTGGGCTCCGGGAGCACTTTGGCTTCGGG
TGGAGCCTGCTTTCCAGCAGCGTGACACGCTTCCCTGGATTGGAGCCAGACTGCTTCCGGGTCACCTG
CCATTGAGAGGCGCAGTCAATCTTACGCTCGAGGCTCTCTGAGTCAAGAAACATTTTCAGACCTATG
GAACTACTTCTGAAAGACGCTTCTGCTCCCTTGGCTCCCAAGCAATGGATGATTGGATGCTGCTCC
CCGAGCATTATGACATGTTCTCACTGAGAGCAGCCAGTCCAGATGAGACTCCAGAGTCCAGAGGCTG
CTCCCTCCCTGGCCCTTGCACAGCAGCTCTCTACACAGCGGCGCTTGCACAGCCCTCTCTGGCCCT
GTCTCTCTCTGCTCTCTCCCTCCAGAGAACCTACAGAGCAGCTACGTTTCTGCTCTGGGCTCTCTGCTCT
GGAGACGCAAGTCTGTACTTGCAGTCT
CTGCTCTCTGCTCTCTGCT
CAGCAGTCAAGCAGCTGAGCGAGGTTTGTGAGGCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCT
GGCTGGCCCT
ACACTTTTGGACATAGT

➤ Total Sequence Length

The full FASTA sequence was concatenated into a single continuous string and its length determined by direct character count.

Total sequence length = 2,512 nucleotides (bp)

➤ Nucleotide Count

Adenine (A) = 532

Thymine (T) = 639

Guanine (G) = 621

Cytosine (C) = 720

Total (A + T + G + C) = 532 + 639 + 621 + 720 = 2,512

➤ GC Content (%)

GC count = G + C = 621 + 720 = 1,341

GC% = $(1,341 / 2,512) \times 100 = 53.38\%$

➤ AT Content (%)

AT count = A + T = 532 + 639 = 1,171

AT% = $(1,171 / 2,512) \times 100 = 46.62\%$

Check: GC% + AT% = 53.38% + 46.62% = 100%

Q4. Data Interpretation

Sequence A: 35% GC

Sequence B: 68% GC

1. Which sequence is more thermally stable?

Seq B (68% GC) is expected to be more thermally stable than Seq A (35% GC).

2. Explanation based on chemical properties of DNA

The G–C base pairs are joined by three hydrogen bonds while the A–T base pairs are joined by only two hydrogen bonds. More hydrogen bonds require more thermal energy to break, so a strand needs more heat to denature as its GC content increases. The G–C pairs also have stronger base-stacking interactions along the helix. Since Seq B has nearly double the GC content of Seq A, its double helix requires a higher melting temperature to denature and making it more thermally stable.

3. Which sequence is more likely to belong to a thermophilic organism?

Seq B is more likely to belong to a thermophilic organism. Thermophiles live in high-temperature environments where the nucleic acids are under constant thermal stress. Elevated GC content is particularly in structural RNAs and often in genomic DNA to help resist heat-induced strand separation and preserves genomic/functional integrity at high temperatures. The GC content of 68% is consistent with the adaptive strategy whereas the AT-rich Seq A (35% GC) is more typical of mesophilic organisms adapted to stable and moderate-temperature environments.

Q5. Practical Activity of NCBI Sequence Retrieval

Accession ID: NC_000932.1 (region 54,958–56,397)

Organism / Source: Arabidopsis thaliana chloroplast genome, complete genome

Sequence length: 1,440 nucleotides (bp)

A / T / G / C counts: A = 404, T = 403, G = 348, C = 285

GC Content: $(348 + 285) / 1,440 \times 100 = 43.96\%$

AT Content: $(404 + 403) / 1,440 \times 100 = 56.04\%$

FASTA Sequence

```
>NC_000932.1:54958-56397 Arabidopsis thaliana chloroplast, complete genome
ATGTCACCACAAACAGAGACTAAAGCAAGTGTGGGTTCAAAGCTGGTGTAAAGAGTATAAATTGACTT
ACTATACTCCTGAATATGAAACCAAGGATACGATATCTTGGCAGCATTCCGAGTAACTCCTCAACCTGG
AGTTCCACCTGAAGAAGCAGGGGCTGCGGTAGCTGCTGAATCTTCTACTGGTACATGGACAACTGTGTGG
ACCGATGGGCTTACCAGCCTTGATCGTTACAAAGGACGATGCTACCACATCGAGCCCGTTCCAGGAGAAG
AAACTCAATTTATTGCGTATGTAGCTTATCCCTTAGACCTTTTGAAGAAGGTTTCGGTTACTAACATGTT
TACCTCGATTGTGGGTAAATGATTTGGGTTCAAAGCCCTGGTGTCTACGTCTAGAGGATCTGCGAATC
CCTCCTGCTTATACTAAAACCTTTCCAAGGACCACCTCATGGTATCCAAGTTGAAAGAGATAAATTGAACA
AGTATGGAGCTGCCCTATTAGGATGTACTATTAAACCAAATTGGGGTTATCCGCGAAAAACTATGGTAG
AGCAGTTTATGAATGTCTACGTGGTGGACTTGATTTTACCAAAGATGATGAGAATGTGAACTCCCAACCA
TTTATGCGTTGGAGAGACCGTTTCTATTTTGTGCCGAAGCTATTTATAAATCACAGGCTGAAACAGGTG
AAATCAAAGGGCATTATTTGAATGCTACTGCGGTACATGCGAAGAAATGATCAAAGAGCTGTATTTGC
CAGAGAATTGGGAGTTCCTATCGTAATGCATGACTACTTAACAGGGGGATTACCGCAAATACTAGTTTG
TCTCATTATTGCCGAGATAATGGCCTACTTCTTACATCCACCGTGCAATGCACGCTGTTATTGATAGAC
AGAAGAATCATGGTATGCACTTCCGTGTACTAGCTAAAGCTTTACGTCATCTGGTGGAGATCATATTCA
CGCGGTACAGTAGTAGGTAACCTTGAAGGAGACAGGGAGTCAACTTTGGGCTTTGTTGATTACTGCGC
GATGATTATGTTGAAAAAGATCGAAGCCGCGGTATCTTTTCACTCAAGATTGGGTCTCACTACCTGGTG
TTCTGCCTGTGGCTTCAGGGGGTATTACGTTTGGCATATGCCTGCTTTGACCGAGATCTTTGGAGATGA
TTCTGTACTACAATTCGGTGGAGGAACTTTAGGCCACCCTTGGGGAAATGCACCGGGTCCCGTAGCCAAC
CGAGTAGCTCTGGAAGCATGTGTACAAGCTCGTAATGAGGGACGTGATCTTGAGTGCAGGGTAATGAAA
TTATCCGTGAAGCTTGCAAAATGGAGTCTGAACTAGCTGCTGCTTGTGAAGTATGGAAGAGATCACATT
TAACCTCCCAACCATCGATAAATTAGATGGCCAAGAGTAG
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Arabidopsis thaliana chloroplast, complete genome
NCBI Reference Sequence: NC_000932.1
[GenBank](#) [Graphics](#)

>NC_000932.1:54958-56397 Arabidopsis thaliana chloroplast, complete genome
ATGTCACCACAAACAGAGACTAAAGCAAGTGTGGGTTCAAAGCTGGTGTAAAGAGTATAAATTGACTT
ACTATACTCCTGAATATGAAACCAAGGATACGATATCTTGGCAGCATTCCGAGTAACTCCTCAACCTGG
AGTTCCACCTGAAGAAGCAGGGGCTGCGGTAGCTGCTGAATCTTCTACTGGTACATGGACAACTGTGTGG
ACCGATGGGCTTACCAGCCTTGATCGTTTGAAGGAGACAGGGAGTCAACTTTGGGCTTTGTTGATTACTGCGC
GATGATTATGTTGAAAAAGATCGAAGCCGCGGTATCTTTTCACTCAAGATTGGGTCTCACTACCTGGTG
TTCTGCCTGTGGCTTCAGGGGGTATTACGTTTGGCATATGCCTGCTTTGACCGAGATCTTTGGAGATGA
TTCTGTACTACAATTCGGTGGAGGAACTTTAGGCCACCCTTGGGGAAATGCACCGGGTCCCGTAGCCAAC
CGAGTAGCTCTGGAAGCATGTGTACAAGCTCGTAATGAGGGACGTGATCTTGAGTGCAGGGTAATGAAA
TTATCCGTGAAGCTTGCAAAATGGAGTCTGAACTAGCTGCTGCTTGTGAAGTATGGAAGAGATCACATT
TAACCTCCCAACCATCGATAAATTAGATGGCCAAGAGTAG

Change region shown

☐ Whole sequence
☒ Selected region
from: 54958 to: 56397 [Update View](#)

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Related information

Interpretation:

This 1,440 bp sequence, retrieved from the Arabidopsis thaliana chloroplast genome (NC_000932.1), corresponds to the *rbcl* gene region, which encodes the large subunit of RuBisCO, the enzyme responsible for the first major step of carbon fixation in photosynthesis. Its GC content (43.96%) is moderate but slightly the AT-biased (56.04% AT), a pattern typical of plant chloroplast genomes, which generally show lower GC content than nuclear genomes. This composition reflects the evolutionary origin of chloroplasts from cyanobacterial endosymbionts and influences codon usage, translational efficiency and gene expression within the organelle. The near-balanced yet AT-leaning composition also means the coding strand is moderately, rather than extremely, stable thermally and consistent with *rbcl* functioning under normal physiological plant temperatures rather than thermal-stress conditions.

Q6. Can two DNA sequences have the same length but different GC contents? Explain with a suitable biological example.

Yes, absolutely. Seq length only indicates the total no. of nucleotides, not the identity or composition. Two seqs can have completely different arrangements and ratios of A, T, G, and C while summing to the same total length.

Biological Example

The human genome averages ~41% GC overall, but regions of identical length differ sharply: CpG islands near gene promoters are GC-rich (often >60%), while intergenic and heterochromatic regions of the same length can be AT-rich (~35–40%). Similarly, comparing a 1,000 bp coding exon of a housekeeping gene to a 1,000 bp intron from the same genome typically reveals different GC content, despite both being exactly 1,000 bp long. This is precisely why GC content is not seq length, is used as a distinguishing genomic signature between species, genomic regions and even individual genes.